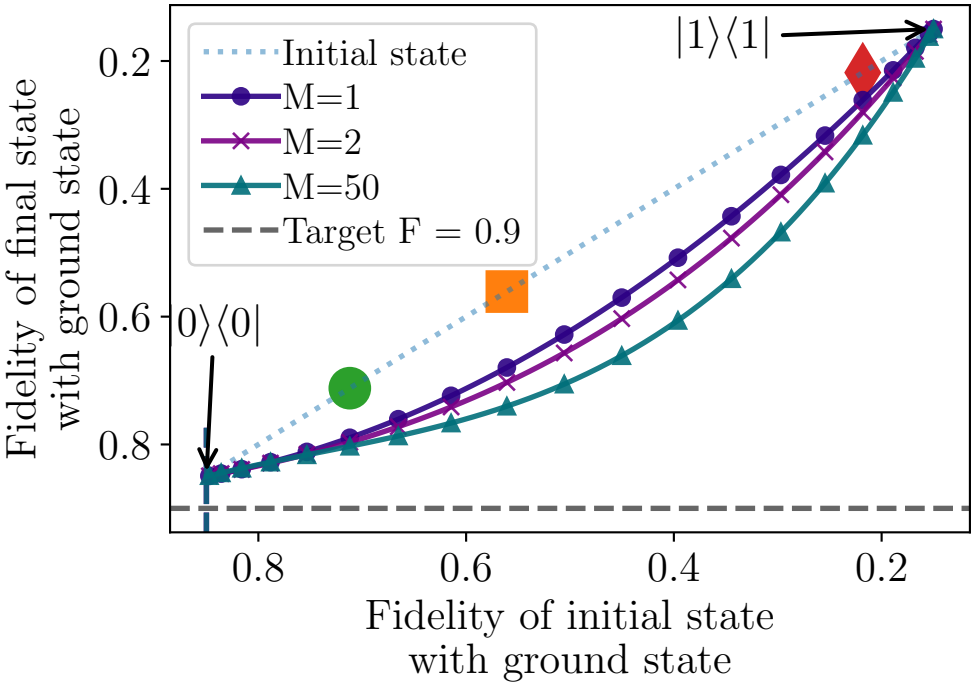

Initial state
M=1
M=2
M=50
Target F = 0.9
$|1\rangle\langle 1|$
$|0\rangle\langle 0|$
Fidelity of final state with ground state
Fidelity of initial state with ground state
0.2
0.4
0.6
0.8
0.8
0.6
0.4
0.2